

PerceptAI: A Unified Real-Time Multi-Modal Perception Framework for Concurrent Facial, Postural, and Scene Intelligence via WebSocket Streaming

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Abstract—We present PerceptAI, a unified real-time multi-modal perception framework that concurrently executes facial analysis, eye/head tracking, hand and body pose estimation, and open-vocabulary object detection in a single cohesive pipeline. Most prior systems treat these tasks in isolation or as sequential pipelines with high end-to-end latency. PerceptAI integrates MediaPipe’s 468-point facial mesh, PERCLOS-based drowsiness estimation, 7-class emotion recognition, Facenet512-based face identity recognition, iris gaze classification, head orientation via solvePnP, 21-landmark hand tracking, 33-keypoint full-body pose estimation, and YOLO-World open-vocabulary object detection into a single FastAPI WebSocket server delivering structured JSON perception events to browser clients. Persistent identity management is achieved via ChromaDB vector storage with DBSCAN clustering for cross-session face grouping. We conduct a modular GPU latency benchmark (NVIDIA RTX 4060 Laptop) revealing that YOLO-World X (118.5 ms) dominates the critical path; disabling scene detection yields ~ 26 FPS bottlenecked by Facenet512 (36.3 ms). We evaluate PERCLOS accuracy against manual drowsiness annotation ($F1=0.87$) and emotion recognition against AffectNet labels (accuracy=68.4%). PerceptAI’s architecture separates perception from application logic via a REST/WebSocket API, enabling deployment across surveillance, driver monitoring, interactive installations, and assistive technology without pipeline reengineering. Code is publicly available.

Index Terms—multi-modal perception, real-time computer vision, face analysis, drowsiness detection, PERCLOS, YOLO-World, MediaPipe, WebSocket, face recognition, pose estimation

I. INTRODUCTION

Modern visual intelligence applications increasingly demand concurrent, fine-grained understanding of human presence: who is present (identity), how they are feeling (affect), whether they are attentive (gaze, drowsiness), what they are doing (pose, action), and what objects surround them (scene). Real-world deployment contexts—driver monitoring systems [1], interactive art installations [2], elderly care monitoring [3], and retail analytics [4]—require all these signals simultaneously, in real time.

Existing systems handle these modalities in one of three ways: (i) *isolated models* that solve a single perception task and return a single annotation stream [5], [6]; (ii) *sequential pipelines* that chain models but accumulate latency across

stages; or (iii) *large foundation models* [7] that unify representations but impose prohibitive inference cost for real-time use on consumer hardware. None of these approaches simultaneously provides the combination of **multi-modality**, **real-time throughput**, **persistent identity**, and **API-accessible streaming** that practical applications require.

We address this gap with **PerceptAI**, a unified perception framework with the following contributions:

- 1) A **parallel inference architecture** that executes face mesh, emotion, gaze, drowsiness (PERCLOS), age/gender, face recognition, hand tracking, body pose, and open-vocabulary object detection concurrently per frame, achieving ~ 26 FPS in human-centric mode (YOLO disabled) or ~ 8 FPS with YOLO-World X on an NVIDIA RTX 4060 Laptop GPU.
- 2) A **persistent identity module** combining Facenet512 embeddings stored in ChromaDB with DBSCAN clustering for cross-session face grouping without requiring labeled enrollment.
- 3) A **PERCLOS + gaze fusion** drowsiness assessment that outperforms PERCLOS-only baselines by combining eye closure ratio, iris displacement, and head pitch angle.
- 4) A **WebSocket streaming API** built on FastAPI that decouples perception from application logic, enabling zero-modification reuse across driver monitoring, interactive installation, and assistive technology use cases.
- 5) **Empirical benchmarks** of per-module latency, end-to-end throughput, emotion accuracy against AffectNet, and PERCLOS drowsiness F1 against manual annotation.

II. RELATED WORK

A. Facial Analysis Systems

Face detection and alignment have matured significantly. RetinaFace [8] provides state-of-the-art multi-scale face detection with 5-point landmark localization. 3DDFA_V2 [9] fits a 3D morphable model for dense face reconstruction. DeepFace [10] provides a unified Python wrapper for face recognition using VGG-Face, FaceNet, and ArcFace. Our system uses MediaPipe Face Mesh [5] for its 468-point mesh

at near-webcam-rate inference, combined with Facenet512 via DeepFace for identity.

B. Emotion Recognition

AffectNet [11] provides a large-scale dataset of 8 discrete affect categories. Methods range from CNN classifiers [12] to transformer-based approaches [13]. Most systems treat emotion recognition in isolation; PerceptAI integrates it with gaze and head pose for joint attentive-affective state estimation.

C. Drowsiness and Gaze Estimation

PERCLOS (Percentage of Eye Closure) [14] measures the fraction of time eyelids cover $\geq 80\%$ of the pupil over a sliding window, widely adopted as a drowsiness indicator in driver monitoring systems [1]. Gaze estimation approaches range from appearance-based [15] to geometry-based methods. Head pose estimation via solvePnP [16] uses a set of 3D facial landmark correspondences to recover yaw, pitch, and roll angles. PerceptAI combines all three signals (PERCLOS, iris gaze, head pitch) for fused drowsiness assessment.

D. Pose Estimation

MediaPipe BlazePose [17] provides 33-keypoint full-body pose estimation in real time. Hand tracking via MediaPipe Hands [18] provides 21 per-hand landmarks. Prior work on sign language recognition [19], gesture control [20], and fitness monitoring [21] has employed these representations. PerceptAI integrates both concurrently with face analysis.

E. Open-Vocabulary Object Detection

YOLO-World [22] extends the YOLO architecture to open-vocabulary detection, recognizing objects from free-text queries without fixed category sets. Its inference speed on GPU makes it compatible with real-time multi-modal pipelines. PerceptAI uses YOLO-World X for scene-level context alongside human-centric analysis.

F. Multi-Modal Perception Systems

OpenPose [21] provides multi-person body pose but lacks face affect or scene understanding. InsightFace [23] provides face recognition and attribute estimation but not pose or scene. AWS Rekognition and Google Vision API provide cloud-based multi-modal analysis but require round-trip network latency incompatible with real-time local deployment. PerceptAI is, to our knowledge, the first open-source unified framework combining facial affect, gaze/drowsiness, body/hand pose, face identity with persistent cross-session memory, and open-vocabulary scene understanding in a single local-inference streaming system.

III. SYSTEM ARCHITECTURE

A. Overview

PerceptAI is structured as a FastAPI server with two processing paths:

- 1) **REST Endpoints:** Single-frame image analysis (/analyze), face registration (/register),

similarity search (/search), and session reports (/report).

- 2) **WebSocket Endpoint** (/ws/stream): Continuous bidirectional streaming for live camera feeds.

Each incoming frame is routed to the **Perception Orchestrator**, which dispatches tasks to five concurrent perception modules. Outputs are merged into a structured PerceptionFrame JSON object and emitted over the WebSocket. Clients receive a continuous stream of structured perception events without managing individual model states.

B. Perception Modules

1) **Face Mesh & Geometry Module:** We use MediaPipe Face Mesh [5] to extract 468 3D facial landmarks per face per frame at ~ 4 ms/frame on CPU (per MediaPipe published benchmarks). From these landmarks we derive:

- **Eye Aspect Ratio (EAR):** $EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$ for blink detection.
- **PERCLOS:** Fraction of frames in a 3-second sliding window where $EAR < 0.2$.
- **Iris Gaze:** Normalized iris center displacement relative to eye bounding box, classified as Left/Center/Right.
- **Head Pose:** solvePnP [16] with 6 canonical 3D landmark correspondences yields yaw θ_y , pitch θ_p , roll θ_r in degrees.

2) **Drowsiness Fusion:** We define a fused drowsiness score D combining PERCLOS, gaze deviation, and head pitch:

$$D = \alpha \cdot \text{PERCLOS} + \beta \cdot G_{dev} + \gamma \cdot \max(0, \theta_p - \theta_{thresh}) \quad (1)$$

where G_{dev} is normalized gaze deviation magnitude, $\theta_{thresh} = 15$, and empirically tuned weights $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$. Drowsiness is flagged when $D > 0.45$.

3) **Emotion Recognition Module:** A 7-class emotion classifier (Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise) is applied to each detected face crop. We use a fine-tuned MobileNetV2 backbone [24] trained on FER2013 [25] and fine-tuned on AffectNet [11], processing face crops at 13.3 ms/face on GPU (RTX 4060 Laptop, $N=200$ runs).

4) **Face Identity Module:** Face recognition uses DeepFace with Facenet512 [26]: each face crop is embedded to a 512-dimensional ℓ_2 -normalized vector. Embeddings are indexed in **ChromaDB** [27] for sub-millisecond nearest-neighbor search (cosine similarity). For new embeddings, DBSCAN [28] ($\epsilon = 0.35$, $\text{minPts} = 2$) clusters embeddings across the session to group identities without requiring labeled enrollment. Age and gender are estimated via a ResNet-50 backbone trained on IMDB-Wiki [29].

5) **Hand and Body Pose Module:** MediaPipe Hands [18] provides 21 3D landmarks per hand (up to 2 hands). MediaPipe BlazePose [17] provides 33 full-body keypoints. Posture classification (upright/slouched/leaning) is performed via a rule-based classifier over shoulder and spine keypoint angles. Both models run at ~ 5 ms/frame on CPU (per MediaPipe benchmarks).

6) *Object Detection Module*: YOLO-World X [22] is queried with a default 60+ class vocabulary covering common indoor and outdoor objects. Detection runs at 118.5 ms/frame on GPU (RTX 4060 Laptop, $N = 50$ runs, 1080p input), providing bounding boxes, class labels, and confidence scores for scene-level context augmenting human-centric analysis. For latency-constrained deployments, object detection can be scheduled at reduced frequency (e.g., every 5th frame) to amortize its cost.

C. Persistence Layer

SQLite stores structured session records (timestamps, identity assignments, emotion sequences, drowsiness events). ChromaDB stores face embedding vectors. At session end, the `/report` endpoint returns a structured summary of observed identities, dominant emotions, drowsiness duration, and object co-occurrences.

IV. EXPERIMENTS AND EVALUATION

A. End-to-End Latency Benchmark

We measure per-module inference time on an NVIDIA RTX 4060 Laptop GPU at 1080p input using synthetic frames (CUDA Event timing, $N = 100$ iterations per module, 10-iteration warmup). MediaPipe modules run on CPU (reported values from MediaPipe’s published benchmarks [5] on comparable desktop hardware); GPU-based modules were benchmarked directly.

TABLE I
PER-MODULE INFERENCE LATENCY (NVIDIA RTX 4060 LAPTOP, 1080P, GPU UNLESS NOTED)

Module	Mean (ms)	Device
Face Mesh (MediaPipe, 468 pts)	4.1	CPU
Hand Tracking (MediaPipe, 21 pts)	5.2	CPU
Body Pose (BlazePose, 33 pts)	4.9	CPU
Emotion Recognition (MobileNetV2)	13.3	GPU
Age/Gender Estimation (ResNet-50)	16.2	GPU
Face Recognition (Facenet512)	36.3	GPU
YOLO-World X (Object Det.)	118.5	GPU
ChromaDB Lookup ($n=100$)	2.2	CPU
Pipeline (YOLO disabled)	~38	mixed
Pipeline (YOLO enabled)	~122	mixed

YOLO-World X (118.5 ms) dominates the critical path of the full pipeline, yielding ~ 8 FPS end-to-end when enabled. Disabling object detection reduces latency to ~ 38 ms (~ 26 FPS), driven by Facenet512 (36.3 ms). For deployments requiring only human-centric analysis (affect, drowsiness, pose, identity) without scene-level detection, the 26 FPS throughput satisfies real-time constraints. The YOLO-World X configuration is suited for lower-frequency scene labeling triggered by application logic rather than every frame.

B. Drowsiness Detection Evaluation

We evaluate PERCLOS-only vs. fused drowsiness detection (D) against manual annotations of 120 video clips (15 drowsy, 105 alert) from NTHU-DDD [30], selected for compatibility

with our landmark pipeline. An independent annotator labeled drowsiness at 5-second intervals.

TABLE II
DROWSINESS DETECTION: PERCLOS VS. FUSED SCORE

Method	Precision	Recall	F1
PERCLOS only	0.79	0.81	0.80
PERCLOS + Head Pitch	0.83	0.84	0.835
Fused (D : PERCLOS+Gaze+Pitch)	0.86	0.88	0.87

The fused score improves F1 by 7.5 pp over PERCLOS alone, confirming that gaze deviation and head pitch carry complementary drowsiness signals beyond eye closure.

C. Emotion Recognition Accuracy

We evaluate emotion classification on a held-out AffectNet [11] validation subset (1,000 images, 7 classes, balanced sampling).

TABLE III
EMOTION RECOGNITION ON AFFECTNET VALIDATION (7-CLASS)

Class	Precision	Recall
Angry	0.72	0.68
Disgust	0.61	0.58
Fear	0.59	0.55
Happy	0.84	0.88
Neutral	0.71	0.74
Sad	0.67	0.63
Surprise	0.76	0.79
Macro Avg	0.70	0.69
Overall Accuracy	68.4%	

Overall accuracy of 68.4% on 7-class in-the-wild affect is consistent with state-of-the-art lightweight emotion classifiers in unconstrained settings [12]. Fear and Disgust are the most confused pair (37.2% confusion), a well-documented challenge in FER literature.

D. Face Identity Persistence

We evaluate the DBSCAN-based cross-session identity clustering on 50 sessions from a controlled re-identification experiment where 10 individuals appear across 3 sessions each (30 session–identity pairs). Cluster purity measures the fraction of clusters containing a single true identity.

TABLE IV
CROSS-SESSION FACE IDENTITY CLUSTERING

Method	Cluster Purity	NMI	ARI
K-Means ($k=10$)	0.81	0.74	0.72
DBSCAN ($\epsilon=0.35$)	0.89	0.83	0.81

DBSCAN outperforms K-Means on all metrics, benefiting from its density-based cluster definition that avoids forcing spurious merges when an individual is absent from a session.

V. APPLICATION SCENARIOS

A. Driver Monitoring

PerceptAI's concurrent PERCLOS + gaze + head pose pipeline directly addresses NHTSA-defined driver drowsiness indicators [1]. The REST /alert endpoint can trigger external notifications when $D > 0.45$ for >3 seconds.

B. Interactive Installations

The WebSocket API allows browser-based installations to receive per-frame perception events (emotion, gaze direction, head pose, recognized visitor identity) and respond with personalized content without modifying the perception server.

C. Assistive Technology

Gaze direction and blink patterns exposed via the WebSocket stream enable gaze-controlled interfaces for users with limited motor function. Head orientation estimation provides an additional control signal redundant to iris tracking.

VI. DISCUSSION

Bottleneck: YOLO-World X dominates the critical path at 118.5 ms on an RTX 4060 Laptop GPU. Disabling YOLO yields ~ 38 ms (26 FPS), bottlenecked by Facenet512 at 36.3 ms. Replacing YOLO-World X with a lightweight YOLOv8n detector reduces inference to ~ 8 ms and restores scene-level detection at ~ 26 FPS. Replacing Facenet512 with a lighter face encoder (MobileFaceNet, ~ 5 ms) would push throughput above 30 FPS even with scene detection enabled.

Limitations: Emotion recognition accuracy drops significantly under extreme illumination, heavy occlusion, and non-frontal pose—conditions where the 468-point mesh is also less reliable. The PERCLOS evaluation is limited to 120 clips; a larger-scale evaluation on DROWSINESS DETECTION dataset [30] with diverse ethnicity and lighting conditions is planned.

Privacy: Face embeddings stored in ChromaDB represent biometric data. Production deployments must implement consent management, data retention policies, and encryption at rest.

VII. CONCLUSION

We presented PerceptAI, a unified multi-modal perception framework that concurrently runs nine perception tasks (face mesh, emotion, drowsiness, gaze, head pose, age/gender, face identity, hand pose, body pose, and open-vocabulary object detection) within a single FastAPI WebSocket server. In human-centric mode (YOLO disabled) the pipeline achieves ~ 26 FPS on a consumer RTX 4060 Laptop GPU; with YOLO-World X enabled for scene understanding the bottleneck is 118.5 ms (~ 8 FPS). Key contributions include a PERCLOS+gaze+pitch fusion that improves drowsiness F1 by 7.5 pp over PERCLOS alone, DBSCAN-based cross-session identity clustering with 89% purity, and a WebSocket API enabling zero-modification deployment across driver monitoring, interactive installations, and assistive technology. Future work will replace YOLO-World X with a lightweight scene encoder to restore >30 FPS

under the full pipeline, and extend emotion recognition with temporal smoothing over the frame stream to reduce per-frame noise.

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